

HEALTHCONNECT

Week 6 Project Status Update

Integration, Advanced Development & Validation

OVERALL STATUS	CURRENT PHASE	NEXT PHASE
ON TRACK	Week 6 – Integration & Validation	Week 7 – Testing & End-to-End Validation

Project Status Overview

HealthConnect has progressed from individual Week 5 deliverables into Week 6 integration, advanced development and validation. Key workstreams are progressing, with ML Engineering successfully validating the integrated pipeline while Data Analytics and Data Science continue cross-track integration and refinement.

Workstream Status

Workstream	Status	Week 6 Progress
Data Analytics	In Progress	Advanced analysis, KPI validation, dashboard refinement and operational recommendations.
Data Science	Ready for Integration	Error analysis completed and Gradient Boosting candidate improved.
ML Engineering	Completed	Reusable integrated pipeline validated; 10/10 tests passed.
Generative AI	In Progress	Assistant evaluation and refinement; integration evidence to be documented.
Project Management	On Track	Integration, dependencies, risks, issues and Week 7 testing coordination.

Key Achievements

- **Advanced Analytics:** Validated key KPIs and analysed booking lead time, previous no-show behaviour, reminders, appointment types, distance and patient segments.
- **Improved Model Performance:** Data Science improved performance from 62.4% to 64.8% accuracy and from 0.677 to 0.687 ROC-AUC.
- **Successful Pipeline Integration:** ML Engineering integrated preprocessing, feature engineering and the candidate model into a reusable pipeline, with 10/10 validation tests passed.
- **Cross-Track Integration:** Data Analytics, Data Science and ML Engineering are exchanging findings, model requirements and technical outputs to support integrated validation.

Current Risks & Considerations

- Moderate predictive performance requires careful interpretation.
- Prediction thresholds need validation against operational capacity.
- Smaller sample sizes may affect some analytical segments.
- Potential data drift and future generalisation should be considered.
- Overbooking should remain within defined capacity controls.

Week 7 Readiness

Week 7 will focus on Testing → Refinement → End-to-End Validation. Planned testing includes Analytics KPI/dashboard testing, Data Science model testing and validation, ML Engineering end-to-end pipeline testing, Generative AI assistant testing, and cross-track end-to-end validation.

Project Management Focus

**Integration • Dependencies • Risks • Issues • Communication • Timeline Tracking • Testing
Coordination**